

Test plan

1. Introduction

This plan defines how the Fin-Cash team will verify that the system meets the Software Requirements Specification (SRS) for Capstone 2025-2026 (Team 13). Fin-Cash is a web-based platform that imports financial data from CSV and XLSX files, generates reports, shows KPI dashboards, forecasts trends and controls access by role (RBAC). The plan covers what is tested, how, with which tools, by whom and by when.

2. Scope

In scope:

- Secure login and Role-Based Access Control (RBAC), including admin management of accounts, roles and permissions
- Importing and validating financial data from CSV and XLSX files
- Processing and aggregating data using predefined calculation rules
- Interactive dashboards showing key financial KPIs
- Automated reports: income statements, balance sheets and cash flow reports
- Financial forecasting from historical data
- Exporting reports and analytics to PDF, XLSX and CSV
- Activity logging (audit log) for security
- Non-functional checks: 3-second dashboard load, TLS/AES-256 encryption, 99.5% uptime, daily backup, browser compatibility (Chrome, Edge, Firefox)

Out scope:

- Direct banking transactions
- Replacing existing accounting systems
- Integration with external legacy databases (a possible future enhancement)
- Testing the corporate IT infrastructure, network and third-party services

3. Test Strategy

Testing follows the five activities in section 6 of the SRS. Each is run against sample data supplied by the project team, and defects are logged and retested after fixing.

- **Functional testing:** verify data import and validation, report generation, forecasting, export and audit logging against the 8 functional requirements, using written test cases.
- **Security and access control testing:** confirm each role (finance staff, analyst, manager/executive, administrator) sees only the features and data allowed, and that unauthorized users cannot export restricted reports.
- **Usability testing:** hands-on sessions with representative users to check dashboard clarity and ease of navigation.
- **Performance and stress testing:** measure dashboard and chart load times with multiple simultaneous users against the 3-second target.

- **Data accuracy verification:** compare generated summaries and exported files with the original source datasets, with zero calculation discrepancy as the pass criterion.
- **Compatibility testing:** run the main flows in Chrome, Edge and Firefox.

4. Environmental Needs

Software requirements:

- Modern web browsers: Google Chrome, Microsoft Edge, Mozilla Firefox
- Microsoft Excel or similar, to prepare and check .XLSX and .CSV test files
- Fin-Cash application build and test database loaded with sample financial data
- Test case file and defect tracking sheet
- Load-testing tool for the multi-user performance tests

Hardware requirements:

- Computers or laptops for the testers, able to run a modern browser
- Server or hosting environment for the Fin-Cash web platform
- Stable internet or internal network (Intranet) connection
- Several devices to simulate concurrent users

5. Staff & Responsibilities

Role	Staff Member	Responsibilities
Test Lead	Mohamed Waleed	Owens the test plan, assigns tasks, tracks progress and writes the final test report
Functional Tester	Marwan hazem	Tests data import, report generation, forecasting and export; writes test cases
Security Tester	Youseef mohamed	Tests login, RBAC, admin permissions and audit logging
Usability and Performance Tester	Sama adel	Runs usability sessions and dashboard load tests, and checks browser compatibility
Data Accuracy Tester	Younna reda	Compares reports and exports with the source data and logs defects
Supervisors	Eng. Mostafa, Eng. Marwa	Guide the team and review the plan and results

6. Deliverables & Scheduling

Deliverables / Millstone	Start date	Duration	End date	Comments
Test plan and test cases file	11 Oct 2026	5 days	15 Oct 2026	Includes sample data setup
Functional and security/RBAC testing	18 Oct 2026	7 days	24 Oct 2026	Defects logged and retested
Usability, performance and data accuracy testing	25 Oct 2026	7 days	31 Oct 2026	Includes browser compatibility
Final test report and review	1 Nov 2026	3 days	3 Nov 2026	Submitted to supervisors

7. Review and approvals

Role	Name	Date	Signature
Test Lead	Mohamed Waleed	10 Oct 2026	<i>Mohamed Waleed</i>
Supervisor	Eng. Mostafa		
Supervisor	Eng. Marwa		